

Define Problem Statements

Every successful software project begins with a clear understanding of the problem it intends to solve. Before identifying requirements or designing a system, it is essential to analyse the current business environment, understand existing challenges, determine their root causes and establish the exact problem that requires a solution. This document formally defines the business problem addressed by the Subscription Waste Detector project.

Purpose of this Document

This document investigates the financial challenges associated with managing recurring digital subscriptions. It examines the current subscription ecosystem, identifies the factors contributing to subscription waste, defines the scope of the identified problem and establishes the foundation for the subsequent requirements analysis and software design activities.

Executive Summary

The rapid adoption of subscription-based business models has transformed the way consumers access digital products and services. Entertainment platforms, productivity software, cloud storage, educational services, fitness applications and numerous other digital solutions are now primarily delivered through recurring monthly or annual subscriptions. While this model provides convenience and flexibility, it also increases the complexity of managing recurring financial commitments.

As users subscribe to more services, subscription payments become distributed across different billing dates, payment methods and transaction histories. Although financial institutions accurately record every payment, they generally present transactions chronologically rather than providing a consolidated view of recurring subscription expenses. Consequently, users often find it difficult to understand their total subscription spending, review active subscriptions or identify services that are no longer required.

This document investigates the underlying causes of these challenges and formally defines the business problem addressed by the Subscription Waste Detector project. Rather than proposing technical implementation, the focus of this document is to establish a clear understanding of the problem that forms the basis for all subsequent software engineering activities.

Document Objectives

- Study the current subscription-based digital environment.
- Identify the financial challenges associated with recurring subscriptions.
- Analyse the underlying causes of subscription waste.
- Formally define the business problem.
- Establish the scope and expected outcomes of the proposed solution.

1. Introduction

The digital economy has undergone a significant transformation over the past decade. Instead of purchasing software or digital content through a single payment, consumers increasingly access products and services through subscription-based pricing models. This shift has enabled service providers to deliver continuous updates while offering customers greater flexibility and lower initial costs.

Today, subscription services are integrated into almost every aspect of daily life. Users commonly subscribe to video streaming platforms, music streaming services, cloud storage providers, productivity software, online learning platforms, fitness applications and professional digital tools. As a result, recurring subscription payments have become an essential component of personal financial activity.

Although the subscription model improves accessibility, it also changes how recurring expenses must be managed. Unlike traditional purchases, subscriptions continue automatically until they are cancelled. Each service may operate using different billing cycles, payment methods, renewal dates and pricing structures, making manual monitoring progressively more difficult as the number of active subscriptions increases.

Managing these recurring financial commitments therefore requires more than simply recording transactions. Users require sufficient visibility into their active subscriptions, recurring expenditure and renewal schedule to make informed financial decisions. Understanding this changing financial environment is the first step toward identifying the business problem that this project aims to address.

2. Evolution of the Digital Subscription Ecosystem

Subscription-based services have expanded beyond entertainment and are now

widely adopted across multiple industries including software, education, cloud computing, digital publishing, professional productivity, healthcare, gaming and communication services. Consumers frequently maintain subscriptions from multiple providers simultaneously, resulting in a growing number of recurring financial commitments that must be managed every month.

Unlike traditional ownership models where payments occur only once, subscription services operate through recurring billing mechanisms. Different providers follow different pricing strategies, renewal frequencies and payment schedules. Consequently, users often manage monthly and annual subscriptions together while using multiple payment sources such as debit cards, credit cards, UPI services and digital wallets.

This continuously expanding subscription ecosystem has significantly increased the complexity of monitoring recurring expenses. Although individual payments remain visible within transaction histories, users often lack a consolidated understanding of their overall subscription commitments, creating the foundation for the financial challenges examined in the following section.

3. Current Industry Scenario

The subscription economy has experienced rapid growth across both consumer and business markets. Services that were traditionally purchased through one-time payments are now commonly delivered using monthly or annual subscription models. Consumers increasingly rely on subscription-based platforms for entertainment, communication, productivity, education, cloud storage, software licensing and various professional services.

This shift has created a recurring financial environment where users may maintain several active subscriptions simultaneously. Each subscription operates independently with different pricing structures, billing dates, renewal periods and payment methods. As the number of subscriptions increases, maintaining complete awareness of recurring financial commitments becomes progressively more difficult.

Although banking applications, payment gateways and digital wallets accurately record completed transactions, they primarily present financial information as chronological payment histories. These platforms generally do not organize recurring subscription payments into a dedicated subscription view, making overall expenditure analysis largely a manual activity.

Current Industry Observations

Area	Observation
Digital Economy	Subscription-based services continue to expand across multiple industries.
Consumer Behaviour	Users frequently maintain multiple active subscriptions simultaneously.
Payment Structure	Recurring payments occur through different billing schedules and payment methods.
Financial Records	Subscription expenses remain mixed with other daily financial transactions.
Financial Visibility	Users often lack a consolidated overview of recurring subscription expenditure.

4. Existing Financial Challenges

The current subscription ecosystem introduces several practical challenges that affect day-to-day financial awareness. These challenges are not caused by subscription services themselves but by the increasing difficulty of organizing and reviewing numerous recurring payments distributed across different financial records.

Challenge 01

Limited Visibility of Recurring Expenses

Transaction histories provide records of completed payments but rarely present subscription expenses as a separate financial category. Users therefore know that payments occurred but often cannot immediately determine the total amount spent on subscriptions during a particular month.

Challenge 02

Automatic Subscription Renewals

Most digital subscriptions renew automatically according to predefined billing schedules. Without regularly reviewing renewal dates, users may continue paying for subscriptions that are no longer actively used or required.

Challenge 03

Fragmented Financial Information

Subscription payments may originate from debit cards, credit cards, digital wallets, UPI applications and other payment platforms. Since these records remain distributed across different payment sources, obtaining a complete subscription overview requires considerable manual effort.

Challenge 04

Difficulty Evaluating Subscription Value

Without organized expenditure summaries, users often find it difficult to evaluate whether each subscription continues to justify its recurring cost. Consequently, unnecessary subscriptions may remain active for long periods without deliberate review.

Business Observation

Collectively, these challenges reduce financial awareness by limiting users' ability to understand recurring expenditure, review active subscriptions and identify opportunities to optimize subscription spending.

5. Why Existing Methods Are Insufficient

Several methods are currently available for monitoring financial transactions. However, these methods primarily record payment activity rather than providing

dedicated subscription analysis. Consequently, users receive transaction information but comparatively little insight into recurring subscription commitments.

Existing Method	Primary Limitation
Bank Statements	Record completed payments without identifying recurring subscription patterns.
Credit Card Statements	Present chronological payment history instead of subscription summaries.
Digital Wallet History	Displays payment activity but provides limited recurring expense analysis.
Manual Tracking	Requires continuous effort and becomes difficult as subscriptions increase.
Personal Memory	Unreliable when managing multiple recurring subscriptions.

The common limitation shared by these approaches is that they focus on recording transactions instead of interpreting recurring financial behaviour. As subscription ecosystems continue expanding, users require analytical support rather than additional transaction records.

6. Root Cause Analysis

The financial challenges discussed in the previous section are symptoms of a broader underlying issue rather than isolated problems. To develop an effective software solution, it is necessary to identify the sequence of events that ultimately leads to subscription waste. Understanding these root causes helps define the actual business problem instead of addressing only its visible consequences.

Increasing Adoption of Subscription-Based Services



Growing Number of Active Digital Subscriptions



Different Billing Cycles and Multiple Payment Methods



Subscription Payments Distributed Across Various Transaction Records



Reduced Visibility of Recurring Financial Commitments



Difficulty Reviewing Subscription Usage and Renewal Decisions



Avoidable Subscription Waste and Reduced Financial Awareness

Root Cause Summary

The primary business issue is not the existence of subscription services or recurring payments. The fundamental problem is the absence of a centralized mechanism that enables users to identify, organize and review subscription-related expenditure separately from general financial transactions. This limitation reduces financial visibility and makes subscription management increasingly difficult as the number of active subscriptions grows.

7. Formal Problem Statement

Based on the analysis presented in the previous sections, the business problem addressed by this project can be formally defined as follows.

Modern consumers increasingly depend on multiple subscription-based digital services. However, existing financial management methods provide limited visibility into recurring subscription expenses because payments are distributed across different billing cycles, payment methods and transaction histories. Consequently, users may find it difficult to identify unnecessary renewals, evaluate recurring expenditure and maintain awareness of their ongoing subscription commitments, leading to avoidable recurring expenses.

8. Need for the Proposed Solution

The problem identified in this document cannot be effectively addressed by simply providing additional transaction records. Instead, users require a solution capable of transforming financial transaction data into meaningful subscription information that supports informed decision-making.

To address this requirement, the **Subscription Waste Detector** has been developed as an AI-powered web application that analyses uploaded transaction data to identify recurring subscription payments, categorize subscriptions, visualize recurring expenditure, generate renewal reminders and provide AI-powered recommendations using the Groq API.

The current implementation is specifically designed to improve subscription visibility and financial awareness. It does not replace banking applications or function as a complete personal finance management platform. Instead, it focuses exclusively on helping users better understand and manage recurring subscription expenses.

Expected Benefits

- Improved visibility of recurring subscription expenses.
- Centralized subscription summaries.
- Better awareness of renewal schedules.
- AI-assisted subscription optimization recommendations.
- Improved financial decision-making.

9. Stakeholder Analysis

The identified business problem primarily affects individuals who maintain recurring digital subscriptions. Understanding stakeholder expectations helps

ensure that the proposed solution remains focused on delivering practical value within the defined project scope.

Stakeholder	Primary Need	Expected Benefit
Individual Consumers	Understand recurring subscription expenses.	Improved financial visibility.
Students	Monitor education and entertainment subscriptions.	Better control of monthly expenses.
Working Professionals	Manage multiple active subscriptions efficiently.	Improved subscription management.
Regular Digital Service Users	Review recurring financial commitments.	Better financial awareness.

10. Project Scope

The project scope clearly defines the capabilities included within the current implementation while identifying functionality that is outside the boundaries of this version of the application.

Included in Current Scope

- CSV transaction upload.
- Excel transaction upload.
- Manual expense entry.
- Recurring subscription detection.
- Subscription categorization.
- Interactive expense visualization.
- Renewal reminders.
- AI-powered recommendations generated using the Groq API.

Excluded from Current Scope

- User authentication.
- Database integration.
- Cloud synchronization.
- Automatic bank connectivity.
- Email and SMS notifications.
- Investment recommendations.
- Loan management.
- Budget management.
- Custom AI model training.

11. Success Criteria

Objective	Success Indicator
Subscription Visibility	Users can clearly identify recurring subscriptions.
Expense Awareness	Monthly subscription expenditure is presented in an understandable format.
Renewal Awareness	Users can review upcoming subscription renewals.
Decision Support	AI-generated recommendations assist subscription review.
Financial Awareness	Users gain better understanding of recurring commitments.

12. Conclusion

This document examined the evolution of the subscription economy, analysed the financial challenges associated with recurring digital subscriptions and identified the underlying factors contributing to limited subscription visibility. The analysis established that the primary business problem is not the availability of financial transactions but the absence of meaningful subscription analysis.

The need for a dedicated solution was therefore established based on the identified gap between transaction recording and subscription intelligence. The Subscription Waste Detector addresses this need by helping users identify recurring subscription expenses, improve financial awareness and make more informed subscription management decisions using AI-powered recommendations generated through the Groq API.

The problem definition, stakeholder analysis, project scope and success criteria established in this document provide the foundation for the next stage of software engineering activities. The subsequent document, **Requirements Gathering and Analysis**, builds upon these findings to identify the functional and non-functional requirements needed to address the business problem defined in this report.

Transition

With the business problem clearly established, the next document focuses on gathering, analysing and documenting the software requirements needed to design and implement the proposed solution.